

GET YOUR GAME ON!

PLACE THAT NUMBER

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INTRODUCTION

Place that number is a whole class activity that builds student understanding of place value and magnitude, as well as estimation skills and overall number sense. Working cooperatively as a class, the objective is to accurately mark as many numbers as possible on a number-line. *Place that number* can be differentiated for students from Foundation to Year 6 by including fewer or more place value slots (see modifying the game).

GAME PLAY

The game as it is currently described is suitable for students in Years 1 and 2. Play begins by the teacher (or gamekeeper) ruling a large number-line (0-100) on a whiteboard and a smaller number-line on a notepad. Students are allocated two 10-sided dice (0-9), corresponding to the number of place value slots in play for the game.

A student volunteer rolls the dice and constructs a two-digit number, such that one dice represents the tens and the other dice represents the ones. For example, rolling a 6 and an 8 would allow the student to construct either the number 68 or the number 86. After consulting with the rest of the class, the student then marks approximately where their number belongs on the number-line. However, importantly, the student is not allowed to record the actual number. On the smaller number-line, the teacher replicates the student's estimate, while also recording the actual number. In this way, the teacher is able to keep track of whether the number-line constructed by students is consistent.

Play continues, with students placing numbers until the teacher indicates an error has been made, whereupon the quantity of numbers correctly placed is counted, representing the class' score for that game.

Although memory is an important aspect of the game, students will generally achieve higher scores the more accurately they estimate the position of the number. After the activity, the teacher should fill in all the



Figure 1. Students work as a team to place 15 numbers on their number line.

numbers associated with the respective marks on the number line. This facilitates opportunities for students to reflect on the number line they constructed.

SUPPORTING THE MATHEMATICS

- Some students will attempt to count along the number line. This strategy generally does not work and tends to result in students placing small numbers too far along the line. The superior strategy is to estimate, and use fractional concepts as benchmarks (e.g., half way along the line should be 50, three-quarters should be 75). The game, therefore, rewards a flexible, relativistic approach to the number line, rather than attempts at 'exact' counting.
- It is important to emphasise that the student group is working together as a team. It is expected that there will be much discussion and conjecture throughout the game, as students attempt to work out the most appropriate spot on the number line to mark their number.
- After the game, get the team to evaluate their performance. Ask what they could have done differently. For example, were two particular numbers placed too close together, or too far apart? Were any specific numbers placed in an inappropriate position, making the rest of the game very difficult? If so, which numbers?
- There are also important strategic aspects of the game which reinforce some of the key mathematical concepts, particularly estimation.
 - Firstly, students may choose to take maximum advantage of the pre-existing benchmarks available (e.g., 0 and 100) when constructing their number line. For example, if an astute student rolls a zero and another number, they will often choose to construct a single-digit number (i.e., place the zero in the tens column), as these numbers can always be placed 'close to zero' and are generally easy for students to recall, which is important later in the game.

- Secondly, students may attempt to construct additional benchmark numbers to aid with future decision making. For example, let us assume that, early in the game, a student rolls two dice which can be constructed to create a number that is very close to 50 (e.g., a 5 and a 1). It might be advisable for the student to create this number (51), rather than its inverse (15), as this number can be placed half-way along the line and be used as a benchmark for the rest of the game.

MODIFYING THE GAME

- Construct a number-line with other orders of magnitude and vary the number of ten-sided dice in play accordingly.

Foundation students: Consider playing with one dice and getting students to place numbers on a number line ranging from 0 to 10.

Years 2 and 3: Three dice and a number line ranging from 0 to 1000.

Years 3 and 4: Four dice and a number line ranging from 0 to 10,000.

Years 4 and 5: Six dice and a number line ranging from 0 to 1,000,000.

Years 5 and 6: Three dice, with students having to construct decimal numbers, with one dice representing the tenths column, once dice representing the hundredths column, and one dice representing the thousandths column. Students would place numbers on a number line ranging from 0 to 1.

- Set a target score for the class (e.g., 10 numbers correct, 15 numbers correct). Alternatively, keep a record of the classes' efforts and encourage the class to beat their previous record. I find that including some form of whole-of-class reward (e.g., eating lunch outside) for either reaching their target or beating their record further energises the game.



Figure 2. Does 36 go here?

- Split the class into two groups, and see which group can record the most numbers before making an error.
- For older students, consider playing in smaller groups and having students play the role of games-keeper.
- For older students, consider exploring the order of fractional numbers through playing with modified dice, and getting students to place numbers on a number line from 0 to 1.